MODULE 1

Introduction to Cyber Criminology

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

By the end of this module, you will think like a cyber investigator - not just a gadget user.

- **What cyber criminology is** - and why it changes how you investigate
- **The digital crime scene** - what every first responder must and must not do
- **Frauds, victims and vectors** - the crime triangle in real cases
- **Types of cyber crimes** - financial, social, technical, and against the vulnerable
- **Threat maps and graphs** - turning phone numbers into a picture of the gang
- **Stages of investigation** - from the 1930 call to the charge sheet

What is Cyber Criminology?

Cyber criminology is the **study of cybercrime, cybercriminals and victims** - and the application of that knowledge to **investigation and prevention**.

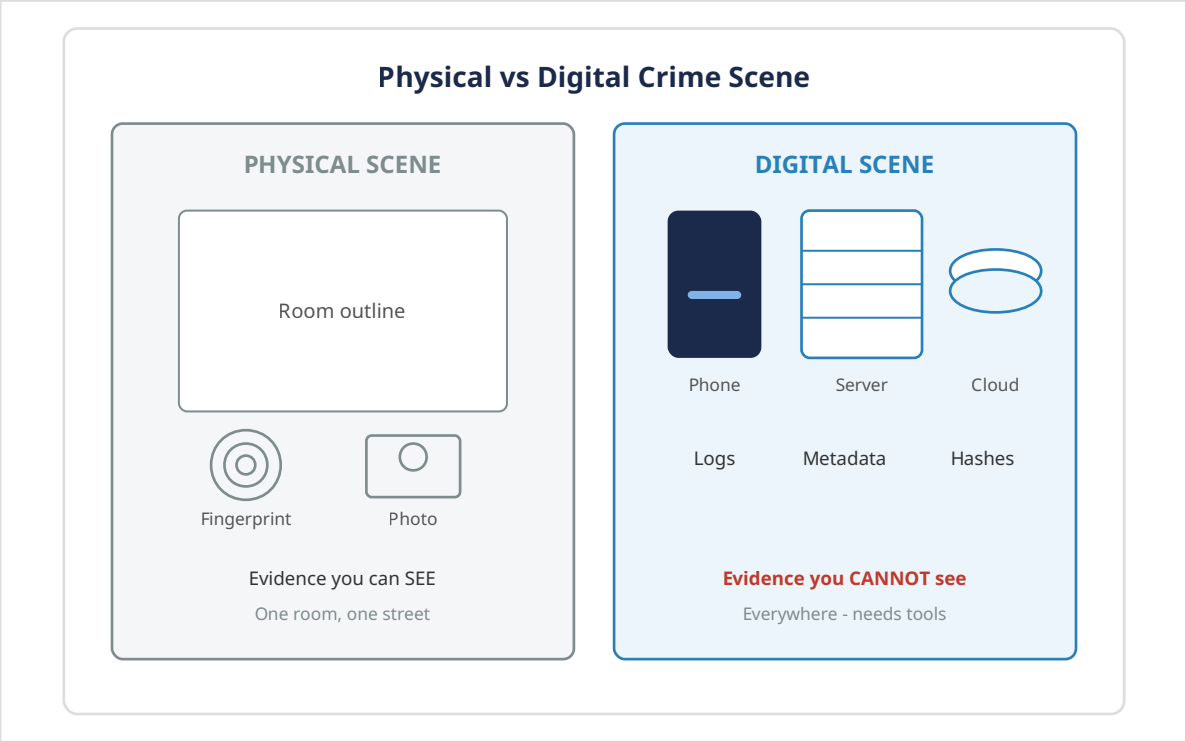
- Classical criminology asks **WHY** crime happens; cyber criminology asks **HOW** it happens through technology
- It connects two worlds: **technology** (how systems, networks and apps work) and **human behaviour** (why criminals offend, why victims fall)
- For you as an investigator it answers four questions: **WHO** did it, **HOW** they did it, **WHO** they targeted - and **HOW TO PROVE IT** in court

Why Every Officer Needs This

Cybercrime is no longer a 'specialist' subject. Almost every case today - fraud, harassment, even murder - leaves a **digital footprint**.

- NCRB 2022: **65,893** registered cybercrimes - a **~24% rise** in one year. Real numbers are far higher
- A phone seized in **any** case can contain the entire evidence chain - if you know how to preserve it
- The **first 24 hours** decide most cyber cases. The trained officer on the spot matters more than the expert lab later
- This course makes YOU that trained officer

Physical vs Digital Crime Scene



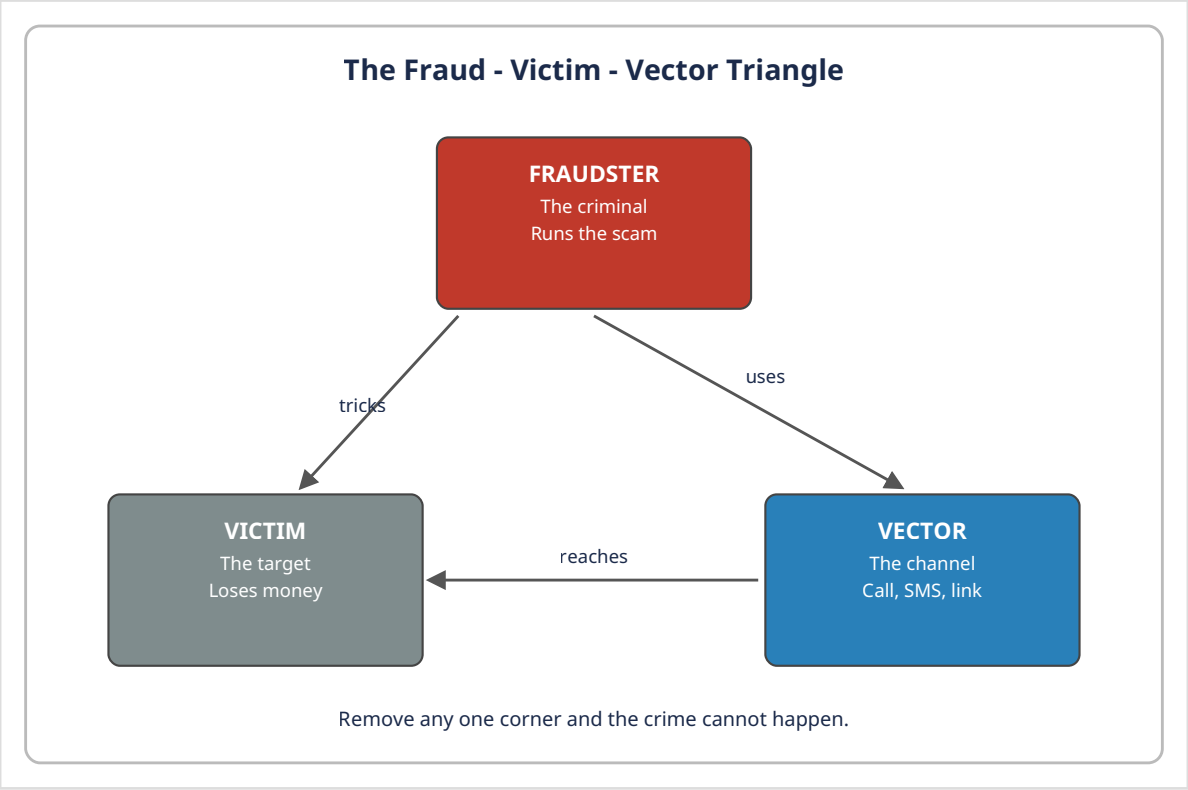
- Physical: **one location** - fingerprints, photographs, evidence you can see and touch
- Digital: **everywhere** - phone, server, cloud. Evidence is logs, metadata and hashes - invisible without tools
- **One wrong click** - switching on a seized phone 'just to look' - can destroy evidence forever

The First Responder's Mistakes

Most digital evidence is destroyed not by criminals, but by **well-meaning officers in the first hour**. Memorise this slide.

- **Switching ON** a seized phone or laptop 'just to check' - changes hundreds of timestamps instantly
- **Not using flight mode** - a remote wipe command can erase the entire phone in seconds
- **Delaying the 1930 call** - stolen money moves in minutes; the golden hour is real
- **Golden rule:** photograph in place, flight mode ON, seize, seal, document - in that order

The Crime Triangle: Fraudster - Victim - Vector



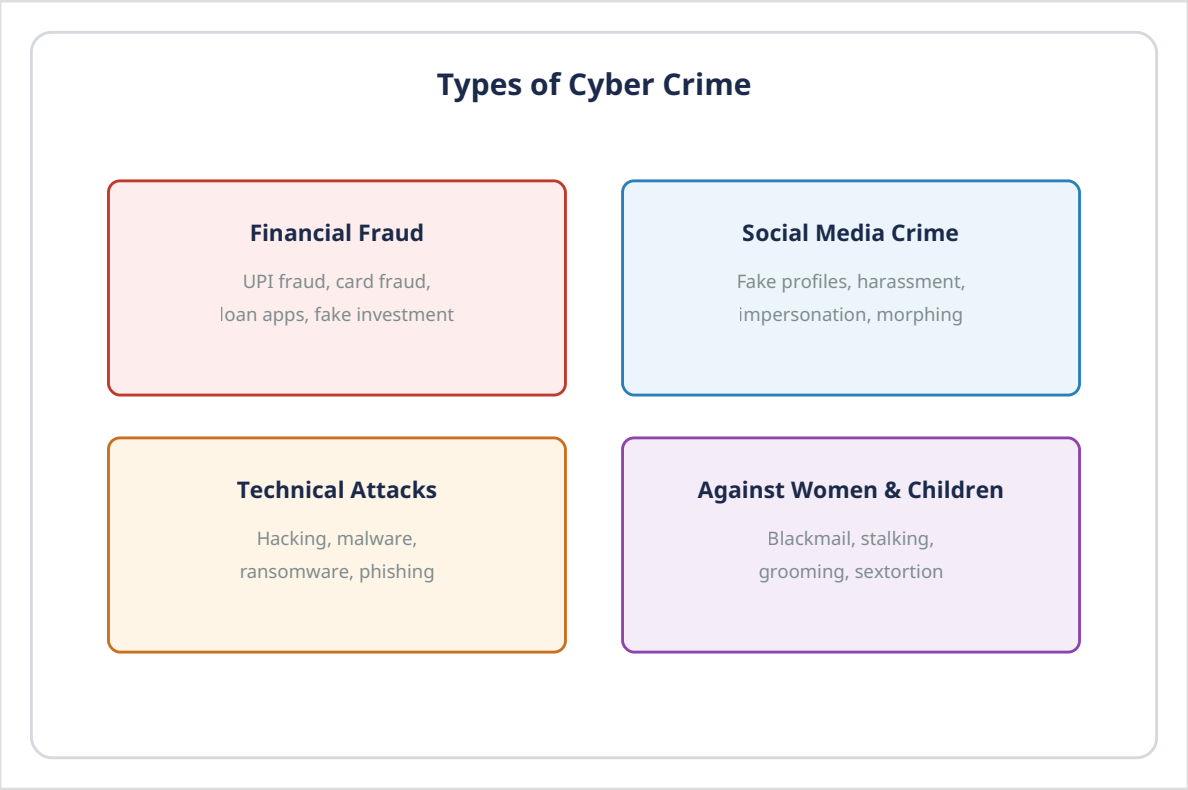
- **Fraudster** - the criminal, often far away, often working in an organised gang
- **Victim** - anyone; trust and fear are the weapons used against them
- **Vector** - the path of attack: a call, a link, an app, a message

Triangle in Action: A Real UPI Fraud

Watch how the triangle fits a case you **will** handle at your station:

- **FRAUDSTER:** a caller from a fake call centre, using a spoofed number and a real bank's name
- **VICTIM:** a retired teacher who trusted the 'bank officer' and shared her OTP
- **VECTOR:** a phone call plus a screen-sharing app - remove any one corner and the crime fails
- Your strategy: **block the vector** (report numbers, freeze accounts via 1930) and **trace the fraudster** (CDR, money trail)

Types of Cyber Crime



Financial Frauds: UPI and OTP

The criminal never 'hacks' the bank. He hacks the **human** - and **you** approve the payment yourself.

- **Fake bank / KYC calls:** 'your account will be blocked' - victim shares OTP, money vanishes in seconds
- **QR code trap:** 'scan to RECEIVE money' - remember: scanning a QR **always SENDS money**, never receives it
- **Screen-sharing apps** (AnyDesk, TeamViewer): a stranger asks you to install one - that is the fraud, right there
- First response: call **1930** immediately, then the victim's bank - every minute counts

Financial Frauds: Investment and Digital Arrest

These frauds run on **manufactured trust** - fake profits, fake uniforms, fake authority.

- **Investment scams:** fake trading apps and Telegram groups show **fake profits** to build trust, then block withdrawals
- **Task scams:** small payments first ('you earned Rs 500'), then a big 'deposit' that vanishes forever
- **Digital arrest:** fake CBI/police video calls threatening arrest - **no such law exists in India**
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> **Tell every victim:** no genuine officer ever demands money on a video call. Ever.

Social Media Crimes

The profile photo is real. The person behind it usually is not.

- **Impersonation:** fake profiles of officers, celebrities or relatives asking for money - verify through a **known number**, never inside the same chat
- **Harassment and non-consensual imagery:** report on cybercrime.gov.in; IT Act ss.67/67A apply
- **Fake news and rumours:** engineered for panic or profit - verify before forwarding; forwarding itself can be an offence
- Evidence tip: screenshot with **URL, date and time visible** - a bare image proves nothing

Technical Attacks

Not every attack needs a genius hacker. Most start with a **leaked password** or an unpatched system.

- **Hacking:** unauthorised access - usually credential theft, not Hollywood-style break-ins
- **Ransomware:** AIIMS Delhi, Nov 2022 - hospital servers encrypted, patient care hit for weeks
- **Data breaches:** BigBasket, Air India (via SITA), Domino's India - assume citizen data is already exposed
- **Botnets:** Mirai (2016) - thousands of hacked CCTV cameras took down major websites

Crimes Against Women and Children

These cases demand **technical skill plus sensitivity** - the victim's dignity is part of the investigation.

- Harassment, morphing, non-consensual imagery - IT Act ss.67, 67A and BNS provisions
- Child sexual abuse material: **POCSO Act** - viewing or forwarding it is a serious offence. Report it; never forward 'as evidence' on WhatsApp
- Preserve fast: screenshots with URL and date-time, plus the victim's own account data download
- A shamed victim stops cooperating - **your behaviour is an investigative tool**

The Criminal's Mind



- Four forces drive cybercrime - **learn them and you can predict the criminal's next move**
- Modern gangs run on **organisation**, not lone hackers in hoodies

How Gangs Actually Operate

Forget the lone hacker. Modern cybercrime is an **industry with departments** - the Jamtara model:

- **CALLERS:** make fraud calls from written scripts, in shifts, with targets and incentives
- **BANKERS:** manage networks of mule accounts that receive the stolen money
- **WITHDRAWERS:** pull cash from ATMs within minutes of the fraud
- Each layer shields the one above it - your investigation must **climb the ladder**, one layer at a time

Why Victims Fall - and What to Tell Them

Victims are not stupid. They are **manipulated by professionals**. Never blame them - a blamed victim stops helping you.

- **Trust:** a uniform on video call, a bank logo - tell them: verify through **official channels**, never the caller's links
- **Fear:** arrest threats freeze thinking - tell them: police **never** arrests anyone on a WhatsApp video call
- **Greed:** double-your-money schemes - tell them: if it sounds too good to be true, it is a trap
- Your first duty to a victim: **stop the bleeding** (1930, bank freeze), then investigate

What is a Threat Map?

A threat map is a **visual picture of who is connected to whom** in a criminal network.

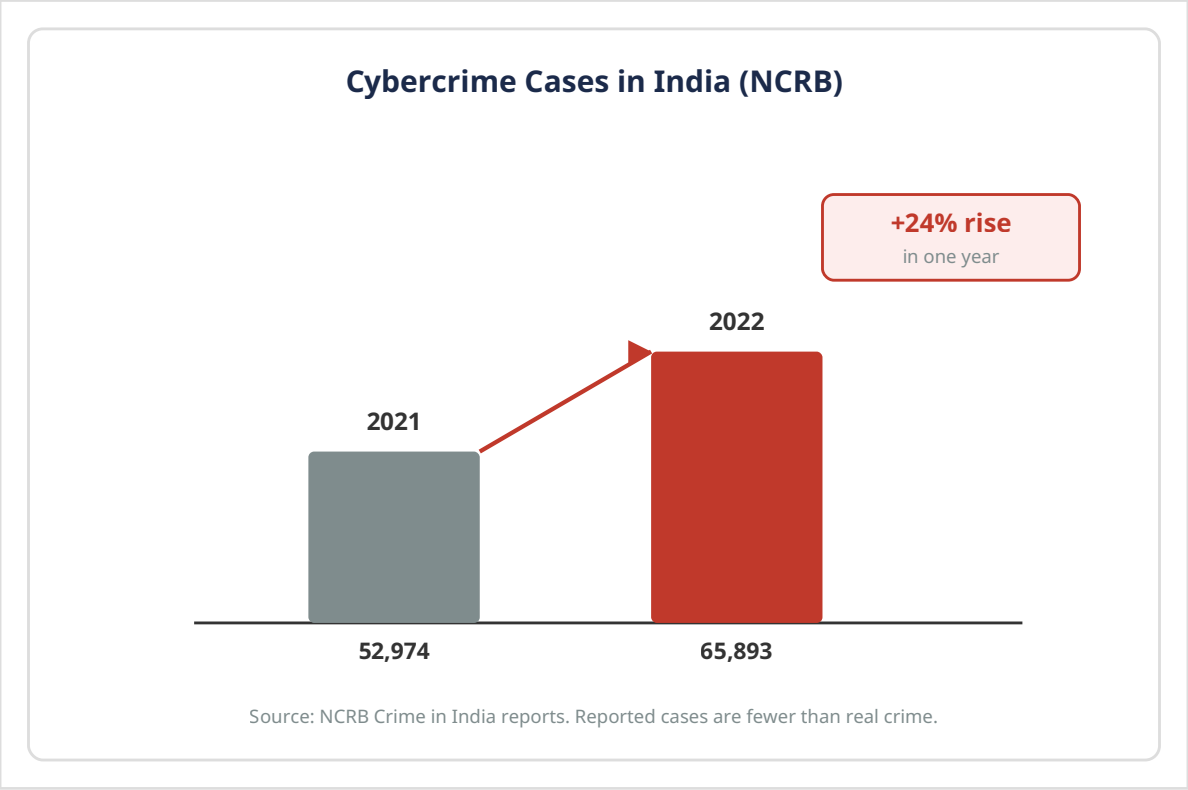
- **Nodes:** phone numbers, bank accounts, devices, people - each one a circle on your map
- **Links:** a call between numbers, a money transfer, a shared device - each one a line
- One fraudster usually controls **many mules** - the map reveals the real head behind the accounts
- Built from CDR, bank statements and device data - with free tools you will learn in this course

Reading a Threat Graph

A link chart turns a pile of phone numbers into **a picture of the gang**. Here is how to read one:

- Find the node with the **most connections** - that is usually the controller. Start your investigation there
- Look for **bridges** - a number connecting two separate groups often belongs to a coordinator
- Money flows **one way** (victim to mule to gang); calls flow **both ways** - direction tells you the role
- If the chart looks like a star with one centre, you have found your prime suspect

Cybercrime Trends in India



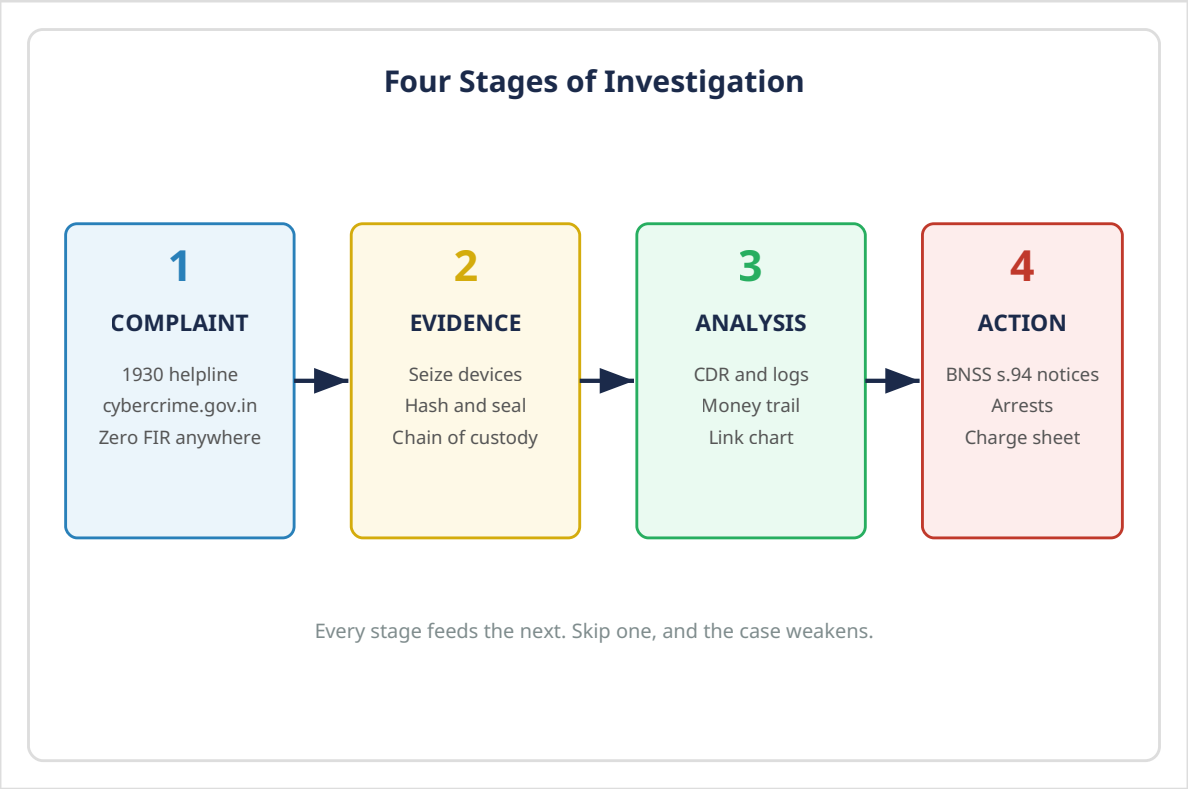
- NCRB 2021 to 2022: **52,974 to 65,893** cases - a **24% jump in one year**
- Reported cases are a **fraction** of real crime - most victims never report
- Financial fraud dominates; digital-arrest and task scams are surging now

What the Numbers Mean for YOUR Unit

Statistics are not trivia. They tell you **where to aim your training**:

- Assume **5 to 10 times** more crime than reported - your station already has these cases, reported or not
- Financial fraud is the bulk - **master UPI and mule-account investigation first**, the rest follows
- Every police station will face these cases. This training is **not optional** anymore - it is the job

Stages of Investigation



- **Every stage feeds the next** - skip one, and the whole case weakens
- Stages 1-2 are won at the **police station**; stages 3-4 in **analysis** - you must master both

Stage 1-2: Complaint and Evidence

A case is **won or lost in the first two stages**. Everything later depends on them:

- **Complaint:** 1930 helpline, cybercrime.gov.in, or any police station - **Zero FIR** means no jurisdiction excuse
- **Evidence:** seize, hash with SHA-256, seal, document - the **chain of custody starts here**, not in the lab
- Golden rule: **if it is not documented, it did not happen** - in court, your register is your witness
- Module 2 covers evidence handling in full detail

Stage 3-4: Analysis and Action

This is where **detective work meets data**:

- **Analysis:** CDR patterns, the money trail, the link chart - follow the data until it names a person
- **Action:** BNSS s.94 notices to banks and telecom companies, arrests, and a charge sheet the court can trust
- A brilliant analysis with weak legal procedure **still loses** - procedure is the bridge from evidence to conviction
- Modules 4 and 5 cover the legal toolkit in depth

Information Management

Investigation **generates** information. Managed well, it becomes evidence; managed badly, it becomes confusion:

- **One case file:** complaint copy, seizure memos, hash values, CDR printouts, bank replies - everything in one place
- **One timeline:** merge every timestamp - call logs, transactions, chats - into a single incident timeline. Judges understand timelines
- **Backup everything:** digital copies of every paper document - paper burns, files corrupt, cases run for years

Real Case: Jamtara - the Phishing Capital

Small towns in Jharkhand became **industrial-scale phishing call centres**:

- Gangs posed as **bank staff**, called victims across India, extracted OTPs and drained accounts
- Structure: **callers** (scripts) + **bankers** (mule accounts) + **withdrawers** (ATMs) - the department model in action
- Lesson: ordinary people + scripts + SIM cards = a fraud industry. **Organisation beats technology**
- Source The Indian Express - reporting on the Jamtara phishing network

Real Case: AIIMS Delhi Ransomware

November 2022 - **India's top hospital** was brought to its knees by cybercrime:

- Ransomware encrypted **5+ servers**; hospital services ran on paper for weeks
- Patient care, research data and administration - all disrupted by one attack
- Lesson: cybercrime is not only about money - it is **national security and public safety**
- Source National media reporting, Nov-Dec 2022

Your Turn: Labs in This Module

Theory without practice is trivia. Your manual's Chapter 1 has **four hands-on labs**:

- **Lab 1:** Draw a link-graph of a fraud network on paper - think in nodes and links
- **Lab 2:** Classify 10 crime scenarios (fraud? hacking? harassment?) - with an answer key
- **Lab 3:** Calculate the NCRB rise percentage and set investigation priorities for your unit
- **Lab 4:** Fraud-victim-vector triangle worksheet - fit real cases into the triangle
- Do them **today** - the next module assumes you can

Key Takeaways

If you remember nothing else from Module 1, remember this:

- Cyber criminology = **technology + human behaviour** applied to investigation
- Every cybercrime has a **fraudster, a victim and a vector** - attack any corner
- Digital evidence is **invisible and fragile** - the first responder decides its fate
- Threat maps turn **data into arrests** - learn to read them
- Crime is rising fast - **trained investigators are the answer**, and that is you
- Never blame the victim - **your behaviour is an investigative tool**

Review Questions

- 1. Define cyber criminology in your own words - and name the four questions it answers for an investigator.
- 2. A seized phone is ringing in the evidence bag. What do you do, step by step, and why?
- 3. Fit a digital-arrest scam into the fraudster-victim-vector triangle.
- 4. In a threat graph, what does the most-connected node usually represent?
- 5. Why are reported NCRB figures lower than real cybercrime? Give two reasons.

NEXT: MODULE 2

Crime Scene Investigation Evidence

Seizing, sealing and chain of custody - where cases are won or lost

Cyber Crime Investigator - Training Manual